

A finite-cotype obstruction for the all-radius entropy version

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Status. Candidate counterexample, likely valid after human review. The result below refutes the finite-cotype converse for the “all $\varepsilon > 0$ ” interpretation of Pisier’s problem on the entropy of $S_n(X)$. It does not solve the “some $\varepsilon > 0$ ” version.

Source problem. Pisier observes that if

$$\log \log N(S_n(X), d, 1 + \varepsilon) = o(n)$$

for some $\varepsilon > 0$, then X does not contain ℓ_∞^n uniformly and hence has finite cotype. He then asks whether the converse holds and asks to characterize the infinite-dimensional Banach spaces satisfying the displayed condition “for some (or all) $\varepsilon > 0$ ” [1, p. 9]. The exact source passage is also TeX lines 570–585 of the parsed source.

then X does not contains ℓ_∞^n ’s uniformly, and hence has finite cotype.

It is tempting to wonder whether the converse holds (recall that $X = \ell_1$ or L_1 are examples of cotype 2 spaces). This question is reminiscent of the conjecture formulated in [19].

Problem. Characterize the infinite dimensional Banach spaces X such that

$$\log \log N(S_n(X), d, 1 + \varepsilon) \in o(n) \quad (n \rightarrow \infty)$$

for some (or all) $\varepsilon > 0$.

Note that this includes all L_p spaces ($1 \leq p < \infty$) by [3, Th. 7.3 and Th. 7.4].

Same question for the class $SQ_n(X)$ of all n -dimensional subspaces of quotients of X . In that case, the $o(n)$ -condition implies that X does not contains ℓ_1^n ’s uniformly, and hence has finite type.

Remark 1.15. Our main result suggests a number of other questions. Among them in the style of

Source theorem used. We use Pisier’s local entropy theorem [1, Theorem 2.1]: for all $1 < r < R$ there is $c = c(r, R) > 0$ such that, for every sufficiently large n and every n -dimensional Banach space E , there is a set $T_n \subset \mathcal{B}_n$ with

$$|T_n| \geq \exp \exp(cn), \quad d(t, u) > r \ (t \neq u), \quad d(t, E) < R \ (t \in T_n).$$

In the parsed source this is Theorem `local1`, TeX lines 606–614.

Theorem 1. *For every $s > 1$ there is an infinite-dimensional Banach space X_s , isomorphic to ℓ_1 , and a constant $a_s > 0$ such that*

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \log N(S_n(X_s), d, s) \geq a_s.$$

Consequently, for every fixed $\varepsilon_0 > 0$, with $s = 1 + \varepsilon_0$, the space X_s has finite cotype but fails Pisier’s entropy condition at the radius $1 + \varepsilon_0$.

Proof intuition. The local theorem says that every finite-dimensional space, in particular ℓ_1^n , has a whole double-exponential cloud of n -dimensional spaces nearby, separated at any prescribed smaller scale. Put all those nearby spaces directly into an ℓ_1 -sum. Because every block is uniformly isomorphic to ℓ_1^n , the whole sum is still just an equivalent renorming of ℓ_1 . But at dimension n the subspace set $S_n(X_s)$ now contains the entire separated cloud.

Proof. Fix $s > 1$. Choose $R > s^2$. Apply the local theorem with $r = s^2$ and center $E = \ell_1^n$. There are constants $c > 0$ and n_0 such that for every $n \geq n_0$ one can choose a finite family

$$\mathcal{F}_n = \{E_{n,\alpha} : \alpha \in A_n\} \subset \mathcal{B}_n$$

satisfying

$$|A_n| \geq \exp \exp(cn), \quad d(E_{n,\alpha}, E_{n,\beta}) > s^2 \quad (\alpha \neq \beta),$$

and

$$d(E_{n,\alpha}, \ell_1^n) < R \quad (\alpha \in A_n).$$

After increasing R slightly, choose for each (n, α) an isomorphism $U_{n,\alpha} : E_{n,\alpha} \rightarrow \ell_1^n$ with

$$\|U_{n,\alpha}\| \|U_{n,\alpha}^{-1}\| \leq R.$$

Define

$$X_s = \left(\bigoplus_{n \geq n_0} \bigoplus_{\alpha \in A_n} E_{n,\alpha} \right)_{\ell_1}.$$

The block diagonal map $\bigoplus_1 U_{n,\alpha}$ is an isomorphism from X_s onto

$$\left(\bigoplus_{n \geq n_0} \bigoplus_{\alpha \in A_n} \ell_1^n \right)_{\ell_1},$$

with distortion at most R . The latter space is isometric to ℓ_1 on a countable index set. Therefore X_s is isomorphic to ℓ_1 , and in particular has cotype 2.

For each $n \geq n_0$, every $E_{n,\alpha}$ is a coordinate subspace of X_s . Thus $\mathcal{F}_n \subset S_n(X_s)$. If a d -ball of radius s contained two distinct members $E_{n,\alpha}$ and $E_{n,\beta}$, the multiplicative triangle inequality for the Banach-Mazur distance would give

$$d(E_{n,\alpha}, E_{n,\beta}) \leq s^2,$$

contrary to the construction. Hence any radius- s cover of $S_n(X_s)$ must use at least one ball for each member of $\mathcal{F}_n$, and so

$$N(S_n(X_s), d, s) \geq |\mathcal{F}_n| \geq \exp \exp(cn)$$

for all $n \geq n_0$. Therefore

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \log N(S_n(X_s), d, s) \geq c.$$

This proves the theorem, with $a_s = c$. □

Remark 1 (Why this is not the some-radius counterexample). *Because X_s is isomorphic to ℓ_1 , the usual Bourgain-Lindenstrauss-Milman embedding theorem cited by Pisier transfers through the global isomorphism and gives the same $o(n)$ double-logarithmic entropy estimate at sufficiently large covering radii. Thus the example is intentionally scoped: it disproves the all- ε finite-cotype converse and shows that the all-radius property is not stable under equivalent renorming, but it does not disprove the some- ε version of the characterization problem.*

Verifier report. No computational verifier is relevant here. The proof has three check points: (1) Pisier’s local theorem is applied with $r = s^2 < R$ and center ℓ_1^n ; (2) the ℓ_1 -sum of uniformly R -isomorphic finite-dimensional ℓ_1 blocks is R -isomorphic to ℓ_1 ; and (3) separation $> s^2$ forces one radius- s covering ball per selected block by the multiplicative triangle inequality. The scope caveat above also checks the apparent tension with the BLM theorem for ℓ_1 .

Novelty check. The run’s lightweight indexes were searched for 1306.5325, `S_n(X)`, `SQ_n(X)`, `Banach-Mazur compactum`, `metric entropy`, and `finite cotype`. No duplicate packet or attempt was found. External exact-phrase searches for the source problem did not reveal a known answer. Novelty confidence is moderate pending human review.

References

- [1] Gilles Pisier, *On the metric entropy of the Banach-Mazur compactum*, arXiv:1306.5325.